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Verbal working memory training in older adults: an investigation of dose response

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ABSTRACT

The WM training protocol proposed by Borella et al. found specific and transfer effects among seniors, however, the studies were carried out in the same socio-cultural context and variations in the procedure were never tested. The present study aimed at analyzing the efficacy of Borella et al.'s training, in terms of short and long-term benefits, in a different socio-cultural context (Study 1), and the effect of change in the training's length (duplicating the number of sessions (Study 2). Participants were randomly assigned to a trained group (N = 18 for Study 1, and N = 23 for Study 2) and active control group (N = 28 for Study 1, and N = 27 for Study 2), and evaluated at pre, post-test and six-month follow-up for verbal WM task (criterion task), and for visuospatial and verbal WM, inhibition, processing speed, executive function, and fluid intelligence measures (transfer tasks). The trained groups had higher performance in all tasks when compared with active control groups after training and at 6 month follow-up. The longer training (Study 2) generated similar gains as the original protocol, with some advantage in far transfer tasks at post-test and follow-up. Study limitations include the small sample sizes. In conclusion, this training was effective in a different socio-cultural context and adding three sessions to the protocol did not significantly change training impact.

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Working memory; cognitive training; older adults; training length; transfer effects

Introduction

Working memory (WM) is a cognitive system, with limited capacity, that is responsible for the maintenance and active processing of the information to be used in complex tasks (Baddeley, 2012). It is very sensitive to aging, in fact, a decline in performance is usually observed in older adults (Bopp & Verhaeghen, 2005, 2007; Borella, Carretti, & De Beni, 2008; Van Geldorp et al., 2015).

Studies focused on WM training have increased in recent years, and these interventions have been shown to be an effective way to increase the WM performance of older adults (Brehmer, Westerberg, & Bäckman, 2012; Morrison & Chein, 2011). The debate in the literature about WM training has revolved around its effect on outcome variables: if the WM training can be generalized to other WM tasks not explicitly practiced in intervention (namely near transfer effect) and to other tasks measuring different constructs but related to WM functioning in older adults, for example: processing speed, attention, and fluid intelligence (far transfer effect) (Li et al., 2008; Morrison & Chein, 2011). The meta-analysis conducted by Karbach and Verhaeghen (2014) pointed out that WM training has a significant impact in tasks explicitly practiced in the training intervention (criterion tasks) and has an important impact in near transfer tasks. For far transfer tasks the effects are less evident. Another recent meta-analysis, conducted by Melby-Lervåg, Redick, & Hulme (2016), confirmed these results.

In this context, studies conducted with a specific verbal WM training procedure (see Borella, Carretti, Riboldi, & De Beni, 2010) produced positive effects replicated over the years (see Borella, Carbone, Pastore, De Beni, & Carretti, 2017a for a synthesis). In the original work of 2010, Borella and colleagues conducted WM training with 40 older adults (65–75 age) randomized into an active control group and a trained group. The objective was to assess the impact of this training on WM measures and evaluate near and far transfer and maintenance effects (8 months). After the intervention, the trained group showed increased performance, when compared with the control group, in the criterion task and near transfer tasks, with eight-month maintenance effects. Far transfer effect was observed, only for the trained group, in all tasks after training, but the results of maintenance were found only for the criterion task, fluid intelligence and processing speed. More recently, the same procedure has shown transfer effects on adults' everyday abilities, assessed with the Everyday Problem Test (Cantarella, Borella, Carretti, Kliegel, & Beni, 2017), as well as on reading comprehension (Carretti, Borella, Zavagnin & De Beni, 2013). The efficacy of this training protocol could be ascribed to different characteristics: 1. it is a verbal training (therefore more favorable for older adults; see Borella et al., 2010), 2. it consists of a limited number of training sessions (3 sessions), 3. it is designed to be challenging (as constant variation in task requirements are introduced during sessions) and use a hybrid procedure (see Borella et al., 2017b, for a discussion).

Although the findings above are encouraging, these studies have been conducted in a high-income country and training data from other regions are lacking. There are

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indeed results in the literature suggesting that low socioeconomic status is associated to lower performance in several cognitive processes, including memory (Rebok, Carlson, & Langbaum, 2007 for a review). Therefore, by replicating the results reported in the original studies, this would enable us to understand whether similar effects can be found in different cultural and socioeconomic contexts (Shipstead, Redick, & Engle 2012).

In addition, dose-response effects have been shown to be a critical issue in cognitive training and currently there is no consensus regarding the ideal WM training length (Karbach & Verhaeghen, 2014; Karr, Areshenkoff, Rast, & Garcia-Barrera, 2014). Therefore, it is also of interest to determine whether the above mentioned procedure generates the same results if training length varies.

Objectives of the current studies

The aim of Study 1 was to replicate the findings from Borella et al. (2010) using the original three-session WM training in the context of a low-income country, in São Paulo, Brazil. Therefore, the objective was to assess the training effects of a three-session individual WM training offered to cognitively intact older adults compared to an active control group.

The aim of Study 2 was to examine the effects of the Borella et al. (2010) WM training protocol offered in 6 sessions instead of 3 to cognitively intact older adults compared to an active control group. The manipulation of the dose-response ratio would contribute to debate on the "ideal" WM training length (Karbach & Verhaeghen, 2014; Karr et al., 2014).

In both studies, the Categorization Working Memory Span task (CWMS) (Borella, Carretti, & De Beni, 2008; Brum, Borella, Carretti, Guidotti, & Yassuda, 2017) was used as the criterion task, as it is similar to the training task. Transfer effects were categorized according to the taxonomy proposed by Noack, Lovden, Schmiedek and Lindenberger (2009), based on a model of human intellectual abilities. For near transfer effects Letter Number Sequencing, Forward and Backward Digit Span test, and Forward and Backward Spatial Span test were used, which tap the same broad ability (WM) but pose different demands as the tests are based on different stimuli and requests from the task used in the training (see Bopp & Verhaeghen, 2005). To assess far transfer effects, we selected tasks which evaluate constructs known to be related to WM process in aging (e.g., de Ribaupierre & Lecerf, 2006), such as Verbal Fluency (to evaluate executive function) (see Borella et al., 2017b), Symbol Search (to evaluate processing speed), Matrix Reasoning (to evaluate fluid intelligence) and the Stroop Color test (to index inhibition).

We expected, on the basis of the Borella's studies and the literature on WM training: 1. that after the intervention the trained group would show a better performance than the active control group for the criterion task and for near transfer tasks (from pre to posttest) and maintenance of gains in the follow up (six months later); 2. for far transfer tasks better performance was expected after the intervention for the trained group that might not be maintenance in the follow up assessment.

Concerning the Study 2, it was expected that the trained group would have higher performance, when compared with the active control group. Most importantly, we expected that the trained group that received six sessions of training would show higher performance in the criterion task, near and far transfer tasks than the three-session protocol as suggested by previous studies (Jaeggi, Buschkuehl, Jonides, & Perrig, 2008; Schwaighofer, Fischer, & Bühner, 2015).

Study 1

Method

Participants

Healthy older adults were recruited to participate in the study on a voluntary basis in a public University of Third Age Program and at a senior center, where seniors go for activities and preventive health care. The inclusion criteria were: age between 60 and 75 years, schooling between four and eleven years, cognitive performance within the expected range for age and education. Exclusion criteria included diagnosis of neurological and psychiatric disorders, not controlled chronic diseases, clinically significant number of symptoms of depression or anxiety, functional impairment, presence of sensory and/or motor impairment that could hinder the participation in training.

To verify the inclusion and exclusion criteria and characterize the sample, the following neurocognitive tests were applied: CERAD Word List to assess episodic memory (Lamberty, Kennedy, & Flashman, 1995; Bertolucci et al., 2001); Clock Drawing Test to evaluated executive functions (Shulman, 2000; Aprahamian, Martinelli, Neri, & Yassuda, 2010); Geriatric Depression Scale (GDS) to measure depression symptoms (Yesavage et al., 1982; Paradela, Lourenço, & Veras, 2005), Geriatric Anxiety Inventory (GAI) to assess anxiety symptoms (Pachana et al., 2007; Massena, de Araújo, Pachana, Laks, & de Pádua, 2015), and Functional Activities Questionnaire to measure functionality (FAQ), (Pfeffer, Kurosaki, Harrah, Chance, & Filos, 1982; Sanchez, Correa, & Lourenço, 2011).

Sixty-seven healthy older adults accepted to participate and after baseline assessment were randomized to study conditions. Participants that missed one or more training sessions or any assessment sessions were excluded from the analyses (See Figure 1). Table 1 describes the characteristics of the final sample ($n = 46$). The trained group

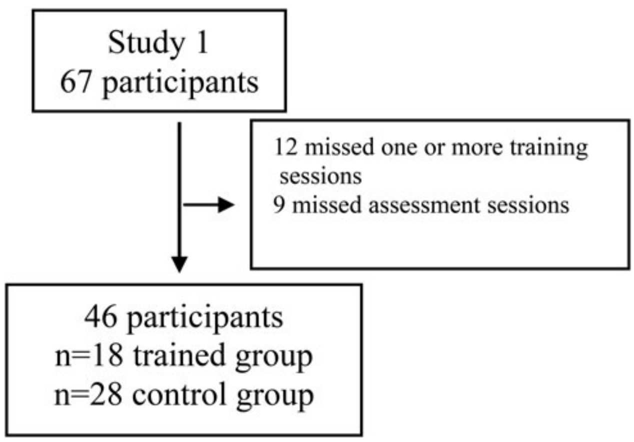

Figure 1. Study 1 design fluxogram.

Table 1. Demographic characteristics (mean and standard deviations) of the Trained group and the Control group participants (Study 1) at baseline.

	Trained group (*N* = 18)	Control group (N = 28)	*p value*	Cohen's d
Age	67.17 (4.40)	66.50 (3.88)	0.60	-0.16
Schooling	9.50 (5.25)	8.46 (4.78)	0.50	-0.21
CERAD delayed recall	5.33 (1.91)	6.21 (1.87)	0.13	0.46
CDT	9.00 (1.13)	8.79 (0.68)	0.47	-0.23
GDS	3.17 (1.38)	3.04 (1.87)	0.78	-0.07
GAI	6.11 (3.99)	5.32 (4.02)	0.51	-0.19
FAQ	0.00 (0.11)	0.00 (0.31)	0.71	-

Note: p value refers to Student *t* test; CDT = Clock Drawing Test, FAQ = Functional Activities Questionnaire, GAI = Geriatric Anxiety Inventory, GDS = Geriatric Depression Scale.

(n = 18) and control group (n = 28) did not differ for age, schooling and performance in the screening tests (see Table 1). Baseline analyses were also run including the participants that dropped out. Results at the pretest did not change (data not shown).

Materials

Criterion task. Categorization Working Memory Span task (De Beni, Palladino, Pazzaglia, & Cornoldi, 1998; adapted to Brazilian Portuguese by Brum, Borella, Carretti, Guidotti, & Yassuda, 2017). The test consists of five sequences of five words, composed of animals and common words. The participants are instructed to memorize the last word of the list, which is followed by a beep sound and, at the same time, to raise the hand when an animal name is heard. At the end of each set of sequences they have to recall the words in the order they were presented. The highest possible score is 20 recalled words and 27 hand raises for animals. The total words recalled in the correct order (CWMS) was considered in this study as the dependent variable.

Near transfer effects measures were Letter Number Sequencing (WAIS-III, Wechsler, 1997; Nascimento, 2004), Forward and Backward Digit Span test (Wechsler Adult Intelligence Scale, WAIS-III, Nascimento, 2004; Wechsler, 1997), and Forward and Backward Spatial Span test (Wechsler Memory Scale III - Wechsler, 1997; Wilde, Strauss, & Tulsky, 2004).

Far transfer effect measures were Semantic Verbal Fluency (animal category) (Brucki & Rocha, 2004), Symbol Search (WAIS-III, Wechsler, 1997, Nascimento, 2004), Matrix Reasoning (WAIS III, Nascimento, 2004; Wechsler, 1997), Stroop Color task (Golden & Freshwater, 1978).

To avoid retest effects, two parallel versions (A and B) of some instruments were used (as described below). The two versions of tests were counterbalanced among participants within each condition at the baseline. At the posttest, participants completed the parallel version and in the follow up they completed the same version used at the baseline.

For Symbol Search the original sequence of pages was used in version A (pages: 1,2,3,4), and in version B the order of pages was modified for (pages: 2,1,4,3). For Spatial Span, Digit Span and Letter Number Sequencing the original sequence was used in version A and in version B the order of sequences was changed. In this study the first sequence became the second and the second sequence became the first, within the same level of difficulty. In the case of CWMS task and Stroop Color task, parallel versions with different items were used. For Matrix Reasoning the even items were used in version A and the odd ones were used in version B. These procedures were adopted following those of previous training studies (e.g. Cantarella et al., 2017; Borella et al., 2010).

Procedures

After the baseline assessment, inclusion and exclusion criteria were verified and if eligible, participants were randomized to the trained group or the active control group according to a block randomization procedure, in blocks of four individuals (Altman & Bland, 1999). Participants completed three assessments: before the intervention (pretest), after the training (posttest) and six months after posttest (follow up). Examiners were blind to participants' experimental condition. Test sessions lasted approximately 90 minutes and were conducted in isolated rooms with ideal conditions for cognitive testing.

Training activities

The trained group participated in three sessions of individual WM training of 30 to 40 minutes each. The sessions occurred in one week on consecutive days (Monday, Tuesday, and Wednesday). The training was the same as described in Borella et al. (2010), adapted for Brazilian language. During training, participants listened to sequences of five words and were asked to recall target words, and raise a hand when they heard an animal name. The number of sequences and words that participants were asked to recall increased during training. The demands of the task also varied and, depending on the session, they could involve having to recall the last or first word in each sequence and words that were followed a beep sound. The processing requirement (hand raising when an animal noun occurred) was also manipulated by varying the frequency of these animal words in the lists. Training sessions were conducted individually. For details of activities carried out in each session please see Appendix 1.

Training sessions were conducted on consecutive days without a two-day interval between sessions, as preconized in the original training protocol. This adaptation was necessary to fit the schedule of the involved institutions and researchers.

The control group was active as participants completed three individual sessions at the same institutions with the same researchers. During the sessions, they answered questionnaires regarding subjective well-being and quality of life (Personal Development Scale (EDEP) (Neri, Cachioni, & Resende, 2002), World Health Organization Quality of Life-Brief (WHOQOL-Brief) (Fleck et al., 2000), Metamemory in Adulthood Questionnaire (MIA) (Yassuda, Lasca, & Neri, 2005) for the same duration as for the trained group. However, in the control group the researcher was in the

Table 2. Descriptive statistics (mean and standard deviations) for the the Trained and Control Groups (Study 1).

	Trained Group			Control Group		
Measure	Pretest	Posttest	Follow-up	Pretest	Posttest	Follow-up
Criterion task						
CWMS	6.11 (2.13)	8.17 (1.88)	7.94 (2.60)	5.75 (1.95)	5.79 (2.33)	5.92 (2.55)
Near transfer						
Letter Number Sequencing	6.28 (2.29)	8.94 (2.79)	8.61 (2.25)	7.32 (2.69)	8.11 (2.93)	8.25 (3.15)
Forward Digit Span	5.39 (2.22)	6.78 (2.13)	7.55 (2.06)	6.04 (2.00)	5.82 (2.00)	6.03 (2.18)
Backward Digit Span	4.22 (1.62)	5.28 (1.56)	4.61 (2.54)	4.68 (1.46)	4.36 (1.52)	4.35 (1.61)
Forward Spatial Span	5.00 (1.74)	5.78 (1.92)	6.38 (1.85)	5.36 (1.49)	5.82 (1.88)	5.42 (1.59)
Backward Spatial Span	4.28 (1.67)	4.94 (1.92)	5.22 (2.07)	4.21 (1.93)	4.00 (1.88)	4.03 (1.45)
Far transfer						
Verbal Fluency	16.78 (5.17)	20.61 (4.91)	20.27 (5.09)	17.93 (4.18)	18.21 (4.96)	18.67 (5.47)
Symbol Search	22.44 (8.08)	23.72 (8.34)	23.05 (8.28)	19.50 (7.75)	22.04 (8.56)	22.14 (7.95)
Matrix Reasoning	4.89 (2.63)	5.50 (2.00)	5.88 (2.21)	4.29 (2.44)	3.79 (2.76)	4.14 (2.33)
Stroop	23.38(17.75)	21.72(14.18)	22.00(12.41)	23.75(20.91)	25.07(17.43)	21.78(17.31)
Stroop Error	1.61 (2.20)	0.89 (1.64)	1.05 (1.39)	1.86 (2.27)	1.75 (2.03)	1.39 (2.04)

Note: CWMS = Categorization Working Memory Span task.

room only to answer doubts about word meaning, and therefore, interaction was minimized. The trained group and the control group completed the assessments according to the same schedule.

This study was carried out according to international ethical standards (Helsinky Declaration) It was approved by the ethical committee of the local University (Process number 141031/2014), and registered in the Brazilian Registry of Clinical Trials (RBR-8H93Y9).

Statistical analyses

For descriptive analyses, the trained group and the control group were compared at baseline with the Student t test (Table 1). To assess training effects, repeated measures ANOVA were implemented with Time (Pretest x Posttest x Follow up) as a within subjects factor and Condition (Trained x Control) as a between subjects factor (see Table 2 for means and SD and Supplementary Material Table 1 for detailed ANOVA results). When interactions were significant, they were decomposed with the Bonferroni multiple comparisons post hoc test. Effect sizes, adjusted for small samples, were calculated (Cohen, 1988; Hedges & Olkin, 1985), and values between 0.20 to 0.49 refer to small effect size, between 0.50 to 0.79 to medium effect and > or equal to 0.80 to large effect size.

Results

Criterion task

For the CWMS the interaction between time and condition was significant. Post-hoc tests showed that for the trained group, between pre to posttest, there was an improvement in performance when compared with the active control group ($p < 0.001$), the same was observed between pretest to follow up ($p = 0.002$). The improvement in performance was not observed for the control group (see Table 2 for means and SD and Supplementary Material Table 1 for detailed ANOVA results).

To ascertain the dimension of the immediate (pre vs posttest) and long-term (pretest vs follow-up) gains the effect size was calculated: only for the trained group it was large at the immediate posttest and became medium at follow-up (see Table 3). In the case of control groups the effects were null.

Near transfer tasks

For Letter Number Sequencing there was an interaction between time and condition: the trained group improved from pre to posttest ($p < 0.001$), and pretest to follow up ($p < 0.001$), the difference between posttest and follow-up was not significant ($p > 0.05$). No differences were observed for the control group (Table 2 and Supplementary Material Table 1). A large effect size for this task between pre to posttest and for pretest to follow up for the trained group was found. For control group the effect sizes were null (see Table 2 for means and SD and Supplementary Material Table 1 for ANOVA results).

For both Forward and Backward Digit Span tasks, the interaction between time and condition was significant; only for the trained group, there was an increase in performance from pre to posttest ($p = 0.003$, $p < 0.001$, respectively) and from pretest to follow up ($p < 0.001$, $p = 0.032$, respectively). The control group did not improve in the three assessment sessions (Table 2 and Supplementary Material Table 1). The effect size was large for Forward Digit Span and medium for Backward Digit Span between pre to posttest for the trained group, but between pre to follow up these results were not maintained (effect size <0.20) (Table 3).

For Forward Spatial Span, scores increased gradually at each assessment in the trained group only, and the difference from pre to follow up reached significance ($p = 0.001$). The control group did not improve between the assessments (Table 2 and Supplementary Material Table 1). Effect sizes indicated that from pretest to posttest and from pretest to follow up the intervention generated a medium effect in this task (Table 3). For Backward Spatial Span the results showed improvement, after the intervention, only for the trained group, between pre to posttest ($p = 0.026$), and between pre to follow up ($p = 0.017$) (Table 2 and Supplementary Material Table 1). The effect size, for the trained group, was medium at the immediate posttest and large at follow-up (Table 3).

Far transfer tasks

The interaction between time and condition showed that for the trained group Verbal Fluency scores increased from pre to posttest ($p = 0.001$) and from pretest to follow up ($p = 0.002$), only for trained group. The performance for the

Table 3. Effect sizes (d of Cohen) for the trained group and control group in Study 1 and Study 2.

	Trained group				Control group			
	Pre to post test gains		Pre to follow up gains		Pre to post test gains		Pre to follow up gains	
Number of sessions	3	6	3	6	3	6	3	6
Criterion tasks								
CWMS	1.00	1.17	0.75	0.88	0.02	−0.01	0.07	0.02
Near transfer								
Letter Number Sequencing	1.03	0.69	0.83	0.34	−0.05	−0.07	−0.16	0.11
Forward Digit Span	0.88	1.01	−0.19	0.09	−0.14	0.25	−0.92	−0.80
Backward Digit Span	0.70	0.55	0.18	0.58	−0.24	−0.02	−0.21	0.05
Forward Spatial Span	0.72	0.69	0.68	0.42	0.13	−0.06	−0.07	−0.06
Backward Spatial Span	0.64	0.63	1.11	0.53	−0.13	−0.18	−0.22	−0.13
Far transfer								
Verbal Fluency	0.74	0.59	0.95	0.45	0.06	0.17	0.19	0.40
Symbol Search	0.53	0.41	0.17	0.27	−0.10	0.08	0.24	0.08
Matrix Reasoning	0.26	0.51	0.40	0.83	−0.19	0.04	−0.06	0.06
Stroop times	−0.10	−0.69	−0.09	−0.59	0.07	−0.03	−0.10	0.02
Stroop Error	−0.36	−0.63	−0.30	−0.34	−0.05	−0.11	−0.21	−0.15

Note: CWMS = Categorization Working Memory Span task; Values between 0.20 to 0.49 refer to small effect size, between 0.50 to 0.79 to medium effect and > or equal to 0.80 to large effect size. Effect size calculation used the pooled SD at baseline.

control group did not change (Table 2 and Supplementary Material Table 1). The effect size was large from pre to posttest and medium between pretest to follow up for the trained group (Table 3).

There was no significant interaction between time and condition for Symbol Search from pre to posttest ($p > 0.05$) and from pretest to follow up ($p > 0.05$) for both groups. But the effect size results showed a medium effect of training in this task from pre to posttest, only for trained group (Table 3). For Stroop, Stroop Error, and Matrix Reasoning tasks there were no significant interactions between time and condition and the effect sizes were null (<0.20).

Discussion

The objective of Study 1 was to replicate the findings from Borella et al. (2010), in a different sociocultural context. For short-term, the results indicated that for the trained group there was a significant improvement in the criterion task (CWMS) and in some of the near (Letter Number Sequencing, Forward and Backward Digit Span, and Forward and Backward Spatial Span) and far transfer measures (Verbal Fluency). These results suggest that this WM intervention may be effective across cultures.

This study included some tests that were similar to the ones used in the original study (Borella et al., 2010) (CWMS, Backward and Forward Digit Span tasks, and Stroop Color Task) and some different tests (Letter Number Sequencing, Forward and Backward Spatial Span, Verbal Fluency, Symbol Search, and Matrix Reasoning). These last tasks were different because the original tests (Dot Matrix task, Cattell test, and Pattern Comparison task) were not validated for Brazilian Portuguese.

The present study included a six-month follow up, in line with Carretti, Mammarella, & Borella (2012), whereas in the original study (Borella et al., 2010) the follow-up was conducted after eight months. We observed that for most variables training gains were maintained after six months (as discussed below).

Criterion task

An improvement was observed, only for trained group, in the criterion task (CWMS) after the training and there was maintenance of this result after six months. The active control group did not change the performance across the assessment phases. Present results are consistent with the original study by Borella et al. (2010) and Borella et al. (2017a). This verbal WM training has a positive impact in the criterion task at posttest even when applied in the older adults living in a different culture. The maintenance of these benefits was described in this study and in previous ones conducted with the same protocol of training (Borella et al., 2010; Carretti et al., 2012; Borella et al., 2014; Cantarella et al., 2017; Borella et al., 2017a, 2017b). The results are also consistent with the meta-analyses showing the efficacy of WM trainings on the criterion tasks, improvement that it is usually maintained at follow-up (Karbach & Verhaeghen, 2014; Melby-Lervåg et al., 2016).

Transfer effects

Training also generated some positive effects on other measures of WM, such as Letter Number Sequencing, Forward and Backward Digit Span, and Backward Spatial Span. These results are not novel as previous studies with this WM protocol have also demonstrated near transfer effects (Borella et al., 2010, 2014, 2017a), they also confirm that such a training procedure allows for a more flexible use of cognitive resources, which might explain the training gains. In fact, the recent meta-analyses have suggested that near transfer effects are commonly found in WM training studies (Karbach & Verhaeghen, 2014; Melby-Lervåg et al., 2016). It is noteworthy that the present WM training protocol was based on verbal WM tasks, but gains were also observed for visual WM measures.

There is debate in the literature about the possibility of observing far transfer effects after WM interventions. On one hand there are some studies and meta-analyses which suggested that it is possible to find effects in abilities not trained (Borella et al., 2010; 2014; 2017b; Cantarella et al., 2017; Karbach & Verhaeghen, 2014), and on the other there are reviews and meta-analyses indicating that it is not feasible to obtain such effects (Melby-Lervåg et al., 2016). Study 1 found some evidences of transfer on Verbal Fluency task, which was maintained at the long term, in agreement with the first series of studies. However, there were no effects on other far transfer tasks (Symbol Search, Matrix Reasoning and Stroop Color Task) in agreement with the studies that question far transfer effects in WM training (Cuenen et al., 2016; Kray & Fehér, 2017; Salminen Mårtensson, Schubert, & Kühn, 2016).

In the original study, Borella et al. (2010) found far transfer effects, which included the Stroop Color task, that was not maintained in the long term, yet, this finding was not replicated in the present study. The means showed that the trained group decreased the time to complete the test, but this decrease did not reach statistical significance at the long-term. This incongruence in findings may be associated with the lack of interval between training sessions in the Brazilian protocol. Yet, this is a hypothesis that should be tested in future studies comparing consecutive training sessions to protocols with intervals.

In sum, findings from Study 1 overall replicated earlier studies carried out in Italy in the context of a low income country such as Brazil. These results are encouraging as they suggest participants in other sociocultural conditions may benefit from a three-session training in the short and long term.

In Study 2, we aimed at investigating whether a longer WM training protocol (six sessions) would generate larger effects in the criterion task and transfer measures immediately after training and at the six-month follow up.

Study 2

An under investigated issue in the cognitive training literature refers to the ideal number of sessions or the ideal training time to maximize training effects. Recently, a meta-analysis with 47 studies (Schwaighofer et al., 2015) investigated factors that may influence near and far transfer effects following WM training. Findings suggested that WM training programs with a higher number of training sessions may generate larger transfer effects measured by visuospatial tasks. However, in another meta-analysis involving 49 studies, Karbach and Verhaeghen (2014) reported the absence of dose-response effects on criterion and transfer measures, in agreement with the study conducted by Karr et al. (2014).

No studies have been found in the literature testing directly dose response effects using the same WM training protocol in older adults. Therefore, we decided to test the effects of the Borella et al. (2010) WM training protocol increasing the number of sessions.

Method

Participants

Sixty healthy older adults accepted to participate in Study 2 and after baseline assessment were randomized to study conditions. Fifty volunteers completed Study 2 (See Figure 2). Recruitment sites, randomization procedures and inclusion and exclusion criteria followed the same procedures described in Study 1.

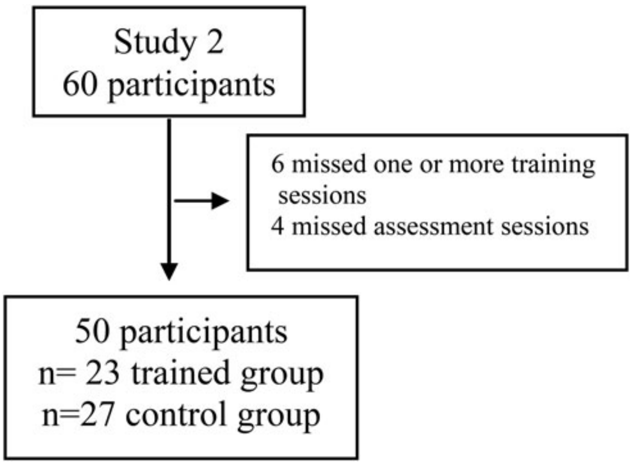

Figure 2. Study 2 design fluxogram.

Table 4 describes the sample characteristics (n = 50). The trained group (n = 23) and control group (n = 27) participants did not differ significantly with respect to age, schooling and scores for screening tests (Table 4). Baseline analyses were also run including the participants that dropped out and results were equivalent (data not shown).

Materials

The same cognitive screening tasks, criterion task, near and far transfer tasks described and used in Study 1 were used in Study 2.

Procedure

Participants, as in Study 1, completed three assessments: pretest, posttest, and six months follow up. Examiners were blind to participants' experimental condition. After pre-test, participants were randomly assigned to the trained or the active control groups, using block randomization based on groups of four individuals, following the same procedures adopted in Study 1.

Training activities

The trained group participated in six sessions of individual WM training of 30 to 40 minutes each. The sessions occurred in two weeks, the first three sessions occurred in consecutive days (Monday, Tuesday, and Wednesday) in the first week and the last three sessions in consecutive days (Monday, Tuesday, and Wednesday) in the second week. The training protocol for sessions 1 to 3 was the same used in Study 1 and other studies (Borella et al., 2010, 2017a). For sessions 4 to 6 the same protocol was applied for a second time, however, the words used in these sessions were reorganized. For this, three different lists were generated: one for the animal words, another for the words that should be memorized, and another for the words that did not need to be memorized. The position of the animal nouns and beep sounds in the sequences was maintained. However, the animal nouns were randomly changed as well as the target (i.e. items to recalled) and non-target words. This was done in order to maintain training engagement. The structure of the training sessions was not modified, only the words were changed without

Table 4. Demographic Characteristics (mean and standard deviations) of the Trained group and the Control group participants (Study 2) at baseline.

	Trained group (N = 23)	Control group (N = 27)	p value	Cohen's d
Age	67.91(3.61)	67.37 (4.41)	0.62	-0.13
Schooling	7.57 (3.34)	7.44 (3.29)	0.89	-0.03
MMSE	27.70 (1.63)	27.48 (1.47)	0.63	-0.14
CERAD delayed recall	5.70 (1.57)	5.67 (1.66)	0.95	-0.01
CDT	8.83 (0.98)	8.74 (0.94)	0.08	-0.09
GDS	3.22 (1.59)	3.19 (1.77)	0.94	-0.01
GAI	5.96 (4.34)	6.52 (3.77)	0.63	0.13
FAQ	0.09 (0.28)	0.15 (0.36)	0.50	0.18

Note: *p* value refers to Student t test; MMSE = Mini-Mental State Examination, CDT = Clock drawing test, GDS = Geriatric Depression Scale, GAI = Geriatric Anxiety Inventory, FAQ = Functional Activities Questionnaire.

Table 5. Descriptive statistics (mean and standard deviations) for the Trained and Control Groups (Study 2).

	Trained Group			Control Group		
Measure	Pretest	Posttest	Follow-up	Pretest	Posttest	Follow-up
Criterion task						
CWMS	6.00 (1.53)	8.83 (2.98)	7.82 (2.44)	6.18 (2.14)	6.15 (2.28)	6.22 (2.43)
Near transfer						
Letter Number Sequencing	7.22 (2.81)	8.91 (2.79)	8.08 (2.64)	6.93 (2.64)	7.52 (2.76)	7.74 (2.66)
Forward Digit Span	4.87 (1.84)	6.52 (2.10)	7.78 (2.52)	4.81 (1.30)	5.19 (1.68)	5.74 (1.50)
Backward Digit Span	3.83 (1.43)	4.04 (1.84)	4.86 (2.00)	3.81 (1.59)	3.89 (1.15)	3.88 (1.33)
Forward Spatial Span	4.96 (1.82)	5.57 (1.61)	5.86 (1.45)	5.00 (1.46)	4.96 (1.50)	5.55 (1.69)
Backward Spatial Span	3.52 (1.31)	4.04 (1.52)	4.52 (1.44)	3.93 (1.38)	3.67 (1.41)	3.88 (1.50)
Far transfer						
Verbal Fluency	15.35 (4.57)	19.22 (4.75)	18.95 (5.13)	16.63 (3.36)	17.22 (3.38)	17.74 (3.89)
Symbol Search	15.96 (5.81)	18.74 (6.33)	18.52 (5.71)	18.41 (5.77)	18.59 (4.83)	19.07 (5.13)
Matrix Reasoning	2.83 (1.61)	3.78 (2.06)	4.52 (1.42)	3.22 (2.08)	3.30 (2.01)	3.33 (1.56)
Stroop	46.21(30.03)	29.95(13.62)	32.95 (9.15)	36.14(32.56)	35.29(28.28)	36.62(26.38)
Stroop Error	3.04 (2.96)	1.48 (1.72)	2.21 (1.75)	2.11 (2.51)	1.85 (2.21)	1.74 (2.23)

Note: CWMS = Categorization Working Memory Span task.

changing the position of the animal words in the lists as well as the words to be remembered.

The active control group participants completed six individual sessions: they were required to fill in questionnaires regarding subjective well-being and quality of life for the same number of minutes as the trained group in the presence of the experimenter, using the similar materials, as for Study 1.

This study was carried out according to international ethical standards (Helsinky Declaration) and it was approved by the ethical board of University (141031/2014), and registered in the Brazilian Registry of Clinical Trials (RBR-8H93Y9).

Statistical analyses

The statistical analyses were identical to those described in Study 1. An additional repeated measures ANOVA (3x2x2) was carried out to evaluate the dose of the training with Time (Pre Test x Posttest x Follow up) as a within subjects factor and Condition (Trained group x Control group) and Dose (tree-session x six-session of training) as between subjects factors.

Results

Criterion task

For the CWMS, analyses showed a significant interaction between time and condition. Bonferroni comparisons highlighted there was an improvement from pre to posttest ($p < 0.001$), and from pretest to follow up ($p = 0.01$) for the trained group. In the control group, no changes were found (Table 5 and Supplementary Material Table 2).

To ascertain the magnitude of the gains, effect sizes were also calculated. The results from pretest to posttest indicated that the intervention, had a large effect for this task only for the trained group, and the same was observed between pretest to follow up (Table 3).

Near transfer tasks

A significant interaction between time and condition was found for Forward and Backward Digit Span task. The Bonferroni comparisons showed an improvement in these tasks, only for trained group, from pre to posttest ($p = 0.002$, $p = 0.001$, respectively) and from pretest to follow up ($p < 0.001$, $p < 0.001$, respectively). The active control group performance did not change across assessment sessions (Table 5 and Supplementary Material Table 2). The effect size, in the case of the trained group, was medium for Forward Digit Span at the immediate posttest, but at long-term the effect was null; the effect was medium for Backward Digit Span task at both the immediate and long-term assessments (Table 3).

For Backward Spatial Span, there was a significant interaction between time and condition. Performance increased from pretest to follow up ($p < 0.001$) in the trained group only (Table 5 and Supplementary Material Table 2), and the effect size was medium for the trained group at the immediate posttest and follow-up (Table 3).

The ANOVA did not show a significant interaction between time and condition for Letter Number Sequence ($p > 0.05$) nor for Forward Spatial Span ($p > 0.05$), but the effect size analyses indicated a medium effect size only between pre and posttest for Letter Number Sequence and for Forward Spatial Span (Table 3), these effect size were not maintained between pretest to follow up (smaller than 0.20, for both tasks).

Far transfer tasks

For Verbal Fluency, there was a significant interaction between time and condition. The Bonferroni comparisons highlighted, only for trained group, an improvement from pre to posttest ($p < 0.001$), and from pretest to follow-up ($p < 0.001$). The active control group maintained the performance across the testing phases (Table 5 and Supplementary Material Table 2). The effect size were medium effect for this task from pretest to posttest for the trained group, which became small at follow-up. At follow-up, a medium effect size emerged for the control group (Table 3).

For Symbol Search there was a significant interaction between time and condition. Post hoc multiple comparisons showed an improvement from pre to posttest ($p < 0.001$) and from pretest to follow-up ($p < 0.001$), only for trained group. The effect size was however small in both cases (Table 3). The active control group maintained a non significant different performance across testing sessions (Table 5 and Supplementary Material Table 2).

For Matrix Reasoning there was a significant interaction between condition and time. Bonferroni comparisons indicated an increase from pre to posttest ($p < 0.001$), and from pretest to follow up ($p = 0.03$) for the trained group, while the active control group was stable during the assessments (Table 5 and Supplementary Material Table 2). There was a medium effect size on this task from pre to posttest and a large effect size from pretest to follow up for trained group only (Table 3).

In the Stroop the interaction between time and condition was significant as well. Post hoc multiple comparisons showed an increase in performance (less time to complete) from pre to posttest ($p = 0.003$) and from pretest to follow up ($p = 0.02$) in the trained group only. No differences were observed for the control group (Table 5 and Supplementary Material Table 2). Effect size analyses indicated a medium effect of training for this task at posttest and follow up (Table 3).

Study 1 and study 2: a comparison

To determine the impact of the training dose, a 3 x 2 x 2 repeated measures ANOVA was performed. The results suggested that there was no significant effect for dose of training for the target and transfer measures (Supplementary Material Table 3).

The effect sizes for the three and six-session trained groups were first qualitatively compared: for the immediate gains (pre-test to posttest), there were comparable effect sizes for the CWMS (large effect), the Forward Digit Span (large effect), the Backward Digit Span (medium effect), the Forward and Backward Spatial Span (medium effect), and the Verbal Fluency (medium effect) tasks. There was a large effect size for Letter Number Sequence for the three-session protocol, whereas the six-session one showed a medium effect size. For Symbol Search a medium effect size was found only for the trained group with three sessions. For Matrix Reasoning a medium effect size was found only for the six-session protocol (Table 3).

For the maintenance effects (pre-test to follow-up), for the three-session trained group there was a large effect size for Letter Number Sequence, Backward Spatial Span, and Verbal Fluency. For the trained group with six sessions a large effect size was observed for the CWMS and Matrix Reasoning. A medium effect size was found for trained group with three sessions for CWMS, Forward Digit Span and Forward Spatial Span; and for the trained group with six sessions for Backward Digit Span, Backward Spatial Span, and Verbal Fluency.

The effect sizes were later transformed into *r* values and three and six session r values were quantitatively compared (Supplementary Material Table 4). The results indicated that the longer training protocol (6-sessions) generated higher immediate gains for Matrix Reasoning, Stroop Test and Stroop Error and higher follow up gains for Matrix Reasoning, Stroop Test and Backward Digit Span in comparison to the three-sessions training protocol. The three-sessions protocol, compared to the 6-sessions, generated higher immediate gains for Letter Number Sequencing and higher follow up gains for Letter Number Sequencing Forward and Backward Spatial Span and Verbal Fluency.

Discussion

In Study 2, the verbal WM training sessions were duplicated to examine the dose response question, regarding specific, near and far transfer effects. The choice of doubling the number of training sessions was made in order to maintain the same structure of the original training, yet offering two opportunities to practice each session. Although longer, the six-sessions training can also be regarded as short, compared to most protocols reported in the literature. A strong point of this WM training is its efficacy despite its short duration. The results for the longer training indicated that for the trained group there was significant improvement in the criterion task (CWMS) and in near (Forward and Backward Digit Span, Backward Spatial Span) and far transfer measures (Verbal Fluency, Symbol Search, Matrix Reasoning, and Stoop) in posttest.

Criterion task

The results of the current study confirm previous findings with a larger dose of training (Borella et al., 2010, 2017a). The trained group improved performance in the criterion task and this benefit was maintained in the follow up, in agreement with the literature (Cuenen et al., 2016; Mitolo, Borella, Meneghetti, Carbone, & Pazzaglia, 2017; Simon, Tusch, Håkansson, Mohammed, & Daffner, 2017). As expected, the active control group did not change the performance during the assessments.

Transfer effects

Study 2 showed near transfer effects that were also maintained at follow-up. Similar results were found in Study 1 and in previous studies reported in the literature (Borella et al., 2010; Kray & Fehér, 2017; Salminen, Mårtensson, Schubert, & Kühn 2016). Results suggest that the Borella et al. (2010) training protocol offered in three or six sessions generates gains in untrained verbal but also visuospatial WM measures.

The results regarding far transfer effects should be viewed with more attention. Study 2 reported, for the trained group only, after six sessions, an improvement for measures of executive function, processing speed, inhibition, and fluid intelligence. These results are in agreement with the original study (Borella et al., 2010) that found far transfer, only for trained group, in fluid intelligence, processing speed, inhibition, and short-term memory with the three-session protocol. Present findings confirm that the six-session protocol can generate gains in cognitive domains not directly involved in training, in agreement with some earlier studies and meta-analyses (Au et al., 2015; Borella et al., 2017a, 2017b; Carretti, Caldarola, Tencati, & Cornoldi, 2014; Heinzel et al., 2014; Schwaighofer et al., 2015). However, they contrast with findings in the literature, for example, Cuenen et al. (2016), Salminen et al. (2016), and Kray and Fehér (2017) who did not find far transfer effects for WM training in older adults.

The follow-up results for the six-session protocol showed maintenance of all far transfer effects after six months without intervention. This contrasts with the original study (Borella et al., 2010) that found maintenance for fluid intelligence and processing speed, but not for inhibition. It also contracts with the findings from Karbach and Verhaeghen's meta-analysis (2014) who concluded that there is a reduction in performance in far transfer measures after months without WM training. It is possible to suppose that this extended far training transfer observed in the Brazilian study is associated to lower cognitive reserve. Observation of mean scores in the Brazilian sample reveals that participants had lower baseline cognitive scores compared to the Italian samples. Thus, it is possible to hypothesize that Brazilian participants may have more room for training gains, which is in line with the compensation hypothesis (Karbach & Verhaeghen, 2014).

Present findings regarding far transfer effects are in agreement with a study conducted by Carretti et al. (2014), who documented maintenance of training effects in far transfer tasks in the follow-up. It is also possible that present results differ from the original study, because the time between pre-test to follow-up was longer for Borella et al. (2010) (eight months vs six months in the current study). However, larger far transfer effects in the present study may be due to the larger dose of training.

General discussion

This article presents the results from two WM training studies using the procedure developed by Borella et al. (2010) but applied in a different socio-cultural context, i.e. healthy Brazilian older adults. In Study 1, we aimed to verify whether it was possible to replicate earlier findings in the context of a low-income country. In Study 2, we aimed to test the impact of this protocol applied with a dose twice as large. The dose-response effect in cognitive training has rarely been investigated directly.

Results from Study 1 revealed that the original WM training protocol generated significant gains and six-month maintenance in the criterion task, in addition to some near and far transfer effects. Study 2 suggested that longer training time produced similar effects (both specific, near and far), maintained after six months as well.

The inspection of effect sizes suggests that the three and six-session protocols generate similar immediate gains for the criterion task, however, maintenance effects seem larger for the longer training. However, statistical comparisons of effect sizes did not support this conclusion.

Study hypotheses predicted larger transfer effects for the longer WM training. For near transfer, gains were, in general, higher for the three-session protocol at follow up. However, for the six-session training, there were higher gains for far transfer effects in the immediate posttest and follow up. These latter results seem to be aligned with Jaeggi et al. (2008) who demonstrated WM training effects on fluid intelligence only after 12 sessions. This effect could not be observed in a lower number of sessions. Taken together, these findings seem to suggest that longer WM training protocols may lead to larger far transfer effects. Thus, the longer training modalities may allow the individual to transpose the training to other skills. In line with this position, Schwaighofer et al. (2015) indicated that longer WM training may have more impact in transfer variables.

General conclusion

To sum up, these two studies evidenced the importance of WM training to improve the performance of older adults in other auditory and visuospatial WM tests not trained. Significant training gains were also observed in tests of processing speed, fluid intelligence, inhibition, and verbal fluency, maintained after six months.

The verbal WM training based on Borella et al.'s procedure has shown positive results, which were replicated in Study 1. This pattern of findings suggests that training effects are not culture specific and that the protocol may be widely used. Study 2 showed that the sessions can be extended with differential findings mainly for some far transfer tasks. As the training protocol is based on recorded auditory stimuli, it may be easily replicated. Another advantage of the verbal WM training is that the duration and number of sessions are short, which facilitate the application of this protocol.

As to limitations of the present study, we may cite that the educational level of participants was similar to the one observed in high-income countries. Therefore, results may not generalize to individuals with lower educational levels. In addition, we highlight that the sample sizes were small, although similar to previous published studies using this training protocol and previous WM training studies with older adults. The small sample size here may justified by the fact that four different groups needed to be composed. Another limitation is the somewhat large number of participants that dropped out of the study. Reasons may be due to the nature of training (individual training with challenging cognitive exercises) or to the high commitment that was necessary (five appointments in two weeks). As directions for future studies, we suggest that they plan to conduct intention-to-treat analyses involving drop out participants.

In sum, we conclude that the investigated WM training protocol generates similar findings when applied in different cultural contexts. Training dose does not seem to impact target and near transfer effects, yet longer WM training protocols may magnify some aspects of far transfer effects.

Disclosure statement

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Appendix 1

Session one has an adaptive procedure, as participants could return to easier levels if they failed to remember the target words and/or raise their hand for animals. This first session was divided in three phases: in the first phase the participant was instructed to recall the last word of each series of words (the words to be memorized vary from two to five), in a second phase, he/she needed to recall the first words in each series (always vary between two to five), and in the third phase he/she need to recall the last words (like the first moment). In all phases the participant need to raise the hand when listened an animal and the participant could orient himself/herself to the target words by a beep sound that was heard after the last word of each sequence of five words. Task difficulty increased if a participant was successful at a given level; in the case of failure, the lowest level was presented. In total, this session included 185 words and 20% were animal words.

In the second session of training the participant needed to raise the hand when he/she listened to an animal name and to memorize the word before the beep sound. But in this session the beep sound could appear in the first or last position or in the middle of the word sequence. The number of words to be recalled varies between two to five. Differently from session one, the number of animal nouns was manipulated: the first two series contained fewer animals words (two animals) when compared with the last two sequences (five animals) in the same level of difficulty. This manipulation was intended to increase the complexity of task's requirements and therefore maintain participants' engagement. In this session participants went through all the levels of difficulty (i.e. the procedure was not adaptive) and the total number of words was 280, including 55% of animal words.

In the third session of training the participant received the same instruction as for the previous ones. In this session the participant had to recall words in the last and in first position on the list, in an alternating manner. The training started with a set of four series of two word lists and in the first series the participant needed to remember the last word of each list, in the second series he/she needed to remember the first word of each list. As for session two this session was not adjusted for difficulty and the total number of words was 280 with 20% of animal words.